

EBN 111/122 ELECTRICITY AND ELECTRONICS

Chapter 1 Basic Concepts

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Chapter 1: Lecture #2 1.5 Power and Energy

- **Power** is the **rate** at which **energy** is consumed or delivered: Joules per second = Watt; [J/s = W].
- Mathematical expression:
$$p = \frac{dw}{dt} = \frac{dw}{dq} \cdot \frac{dq}{dt} = vi$$
- **Instantaneous power** = the rate at which energy is consumed at a specific point in time:
- DC systems: $p(t) = P = VI = \text{constant}$
- AC systems: $p(t) = v(t)i(t) = \text{time function}$

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1.5 Power and Energy

Passive Sign Convention

$p = +vi$

If the reference current direction enters the positive terminal: $p = +vi$

$p = -vi$

If the reference current direction enters the negative terminal: $p = -vi$

$p > 0$: element is absorbing or consuming energy

$p < 0$: element is delivering or supplying energy

Calculate the power being absorbed or delivered (effect of changing signs)

2 A
 $+$
 4 V
 $-$
 $p = +vi$
 $= + (4)(2)$
 $= 8 \text{ W}$
 $\therefore p > 0$:
 $\therefore$ Absorbs 8 W

2 A
 $-$
 4 V
 $+$
 $p = -vi$
 $= - (-4)(2)$
 $= +8 \text{ W}$
 $\therefore p > 0$:
 $\therefore$ Absorbs 8 W

-2 A
 $+$
 4 V
 $-$
 $p = -vi$
 $= - (4)(-2)$
 $= +8 \text{ W}$
 $\therefore p > 0$:
 $\therefore$ Absorbs 8 W

-2 A
 $-$
 4 V
 $+$
 $p = +vi$
 $= + (-4)(-2)$
 $= +8 \text{ W}$
 $\therefore p > 0$:
 $\therefore$ Absorbs 8 W

1.5 Power and Energy

- The law of **conservation of energy** implies that in an electrical circuit $p_{\text{delivered}} = p_{\text{absorbed}}$, or:

$$\sum p = 0$$

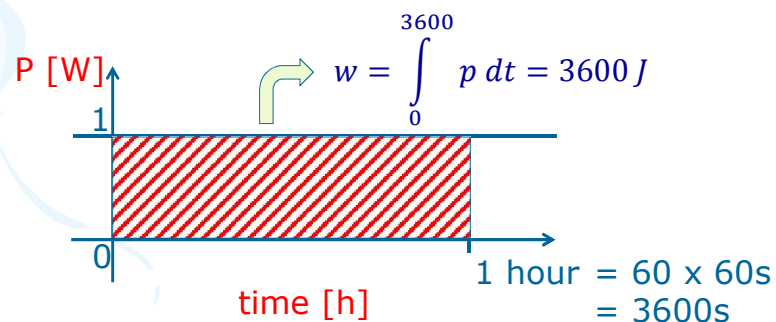
- Energy** is the capacity to do work, measured in joules (J) or alternatively in Watt-hour (Wh).
 - 1 Wh = 3600 J = 3.6 kJ
 - 1 kWh = 3.6 MJ

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1.5 Power and Energy

- Energy absorbed or delivered by an element between times t_0 and t_1 is:

$$w = \int_{t_0}^t p dt = \int_{t_0}^t v i dt$$

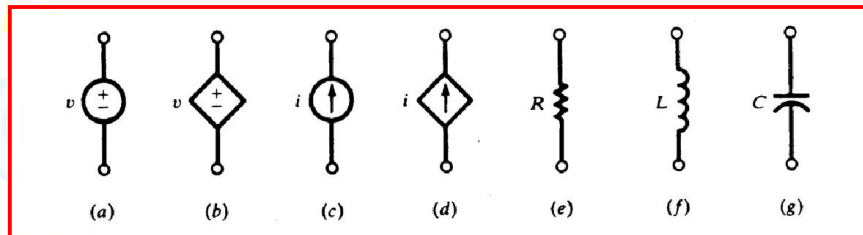


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1.6 Circuit Elements (1)

Active Elements (Energy Sources)
Generators, batteries, op-amps

Passive Elements
Resistors, Inductors,
Capacitors



Independent sources Dependant sources

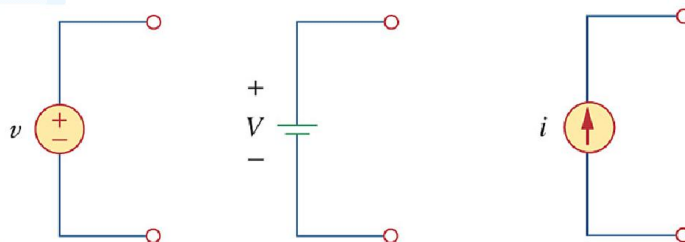
- A dependent source is an active element in which the source quantity is controlled by another voltage or current.
- They have four different types: VCVS, CCVS, VCCS, CCCS. Keep in mind the signs of dependent sources.

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Ideal Independent Sources:

An active element that provides a specified voltage or current that is completely independent of other circuit elements. **ROUND** in shape.

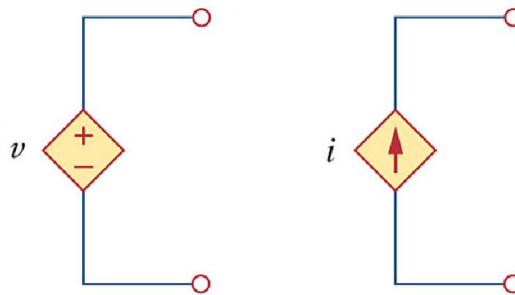
I.e. If $v = 10\text{ V}$ it stays 10 V no matter what!
If $i = 5\text{ A}$ it stays 5 A no matter what



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Ideal **D**ependent Sources:

- An active element in which the source quantity is controlled by another voltage or current. DIAMOND in shape.
- Useful in modeling elements such as transistors, operational amplifiers and integrated circuits.

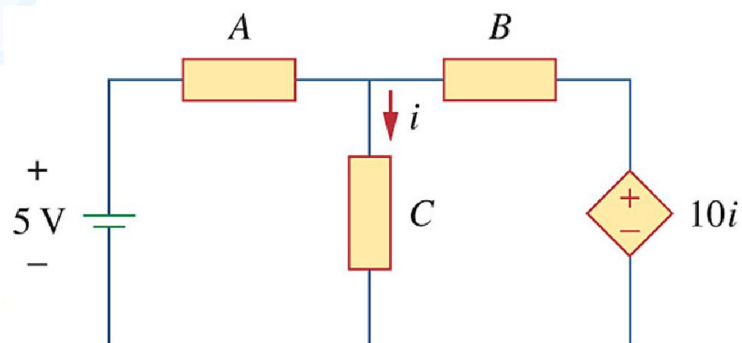


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Types of **D**ependent Sources:

Four possible types of dependent (or controlled) sources:

- VCVS = Voltage Controlled Voltage Source
- CCVS = Current Controlled Voltage Source
- VCCS = Voltage Controlled Current Source
- CCCS = Current Controlled Current Source



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